

Power Plant Engineering
Assignment
Max Marks 15
A batch

Each student shall answer the question assigned to him.

1(a)

The input output curve of a 10MW power station is expressed as follows:

$$i = 10^6(8 + L + 0.4L^2)$$

where i is in kcal per hour and L is in MW.

Determine the average heat rate of the power station for a day if it operates at its full capacity for 12 hours and is kept running at zero load for remaining 12 hours. Also calculate the saving per kWh of energy produced if the energy is generated at a constant 24 hours load (100% load factor).

(b) A common load is to be shared by two power plants. One power plant is a base load plant with 30 MW installed capacity and other power plant is a standby plant with 20 MW capacity. The yearly output of the base load plant is 130×10^6 kWh and that of standby plant is 9×10^6 kWh. The peak load taken by the standby plant is 15 MW and this plant works for 2800 hours during the year. The base load plant takes a peak of 25 MW. Determine the following for both plants:

- (a) Annual load factor
- (b) Plant use factor
- (c) Capacity factor.

2(a)

The maximum (peak) load on a thermal power plant of 60 MW capacity is 50 MW at an annual load factor of 50%. The loads having maximum demands of 25 MW, 20 MW, 8 MW and 5 M are connected to the power station. Determine : (a) Average load on power station (b) Energy generated per year (c) Demand factor (d) Diversity factor.

(b) The capital cost of a power generating equipment in a steam power plant is Rs 80×10^6 . The useful life of the plant is 30 years and its

salvage value is 5% of the capital cost. Determine by the sinking-fund method the amount of money to be saved annually for replacement if the yearly rate of compound interest is 6%.

3 (A)

A power station supplies the following loads to the consumers:

Time (hours)	0-6	6-10	10-12	12-16	16-20	20-22	22-24
Load (MW)	30	70	90	60	100	80	60

- (a) Draw the load curve and estimate the load factor of the plant.
 (b) What is the load factor of a standby equipment of 30 MW capacity if it takes up all loads above 70 MW? What is its use factor?

(B) A plant costing Rs 125,000 has a useful life of 20 years. Find the amount which should be saved annually to replace the equipment at the end of that time (a) by the straight-line method, and (b) by the sinking-fund method if the annual rate of compound interest is 7%. Assume that the salvage value of the equipment is Rs55000.

4.(A)

A power station has to meet the following demand :

Group A : 150 kW between 8 A.M. and 6 P.M.

Group B : 350 kW between 6 A.M. and 10 A.M.

Group C : 350 kW between 6 A.M. and 10 A.M.

Group D : 50 kW between 10 A.M. and 6 P.M. and then between 6 P.M. and 6 A.M.

Plot the daily load curve and determine (i) diversity factor (ii) units generated per day (iii) load factor.

(B) A chimney is 29 metres high and the temperature of hot gases inside the chimney is 320°C. The temperature of outside air is 27°C and furnace

as supplied with 15 kg of air per kg of coal burnt. Calculate

(a) Draught in mm of water.

(b) Draught height in metres of hot gases.

5

At the end of a power distribution system, a certain feeder supplies three distribution transformers, each one supplying a group of customers whose connected load are listed as

follows :

Transformer 1

Transformer 2

Transformer 3

General power
service and lighting

Residence lighting

Store lighting and power

a : 22 H.P., 14kW

e : 15 kW

i : 25 kW

b : 13 H.P., 14kW

f : 12 kW

l : 16 kW

c : 14 H.P.

g : 6 kW

j : 28 kW, 7 H.P.

d : 7 H.P., 12 kW

h : 12 kW

k : 16 kW, 23 H.P.

Demand factor for all type of load is .62 & Diversity factor as 1.35

Use the factor and predict the maximum demand on the feeder. The H.P. load is motor load and assume an efficiency of 75%.

6.(a)

The maximum (peak) load on a thermal power plant of 60 MW capacity is 50 mW at an annual load factor of 50%. The loads having maximum demands of 25 MW, 20 MW, 8 MW and, 5 MW are connected to the power station.

Determine: (a) Average load on power station (b) Energy generated per year (c) Demand factor (d) Diversity factor.

6(b)

In a steam power plant the capital cost of power generation equipment is Rs. 25×10^5 . The useful life of the plant is 30 years and salvage value of the plant to Rs. 1×10^5 . Determine by sinking fund method the amount to be saved annually for replacement if the rate of annual compound interest is 6%.

7

(a)

Two lamps are to be compared:

(a) Cost of first lamp is Re. 1 and it takes 100 watts.

(b) Cost of second lamp is Rs. 4 and it takes 60 watts.

Both lamps are of equal candle power and each has a useful life of 100 hours. Which lamp will prove economical if the energy is charged at Rs. 70 per kW of maximum demand per year plus 5 paise per kWh? At what load factor both the lamps will be equally advantageous?

(b) The input-output curve of a 150 MW station is expressed by the formula,

$$I = 10^6 (250 + 5L + 0.03L^3) + L^4$$

where I is in kcal per hour and L is in mega-watts. Find the load at which minimum heat rate occurs and check with the plot.

8(A)

A power station has to supply load as follows:

Time (hours)	0-6	6-12	12-14	14-18	18-24
Load (MW)	30	90	60	100	50

(a) Draw the load curve. (b) Draw the load duration curve. (c) Select suitable generating units to supply the load. (d) Calculate the load factor. (e) Calculate the capacity of the plant and the plant capacity factor.

(B) The capital cost of a power generating equipment in a steam power plant is Rs 80×10^6 . The useful life of the plant is 30 years and its salvage value is 7% of the capital cost. Determine by the sinking-fund method the amount of money to be saved annually for replacement if the yearly rate of compound interest is 6.5%.

9 (a)

The incremental fuel costs for two generating units A and B of a power plant are given by the following relations,

$$dF_a/dP_a = 0.065 P_a + 25$$

$$dF_b/dP_b = 0.08 P_b + 20$$

where F is the fuel cost in rupees per hour and P is the power output in MW. Estimate

(a) the economic loading of the two units when the total load supplied by the power plants is 200 MW, (b) the loss in fuel cost per hour if the load is equally shared by both units.

(b)

In a steam power plant the capital cost of power generation equipment is Rs. 25×10^5 . The useful life of the plant is 30 years and salvage value of the plant to Rs. 1×10^5 . Determine by sinking fund method the amount to be saved annually for replacement if the rate of annual compound interest is 6%.

10 (a)

The following loads are connected to a power plant:

Type of load	Max. demand (MW)	Diversity factor	Demand factor
Domestic	15	1.25	0.70
Commercial	25	1.20	0.90
Industrial	50	1.30	0.98

If the overall diversity factor is 1.5, determine (a) the maximum load, and (b) the connected load of each type

(b) The incremental fuel costs for two generating units A and B of a power plant are given by the following relations,

$$dF_a/dP_a = 0.06 P_a + 35$$

$$dF_b/dP_b = 0.09 P_b + 25$$

where F is the fuel cost in rupees per hour and P is the power output in MW. Estimate

(a) the economic loading of the two units when the total load supplied by the power plants is 150 MW, (b) the loss in fuel cost per hour if the load is equally shared by both units.

11 (a)

A power station has to meet the following demand :

Group A : 200 kW between 8 A.M. and 6 P.M.

Group B : 100 kW between 6 A.M. and 10 A.M.

Group C : 50 kW between 6 A.M. and 10 A.M.

Group D : 100 kW between 10 A.M. and 6 P.M. and then between 6 P.M. and 6 A.M.

Plot the daily load curve and determine (i) diversity factor (ii) units generated per day (iii) load factor.

(b)

12

At the end of a power distribution system, a certain feeder supplies three distribution transformers, each one supplying a group of customers whose connected load are listed as

follows :

Transformer 1

Transformer 2

Transformer 3

General power
service and lighting

Residence lighting

Store lighting and power

a : 10 H.P., 5kW

e : 5 kW

i : 20 kW

b : 7.5 H.P., 4kW

f : 4 kW

l : 4 kW

c : 15 H.P.

g : 8 kW

j : 10 kW, 5 H.P.

d : 5 H.P., 2 kW

h : 15 kW

k : 8 kW, 25 H.P.

Demand factor for all type of load is .60 & Diversity factor as 1.25

Use the factor and predict the maximum demand on the feeder. The H.P. load is motor load and assume an efficiency of 72%.

13 (A)

Example 3.11. A power station has to meet the following demand :

Group A : 450 kW between 8 A.M. and 6 P.M.

Group B : 250 kW between 6 A.M. and 10 A.M.

Group C : 250 kW between 6 A.M. and 10 A.M.

Group D : 100 kW between 10 A.M. and 6 P.M. and then between 6 P.M. and 6 A.M.

Plot the daily load curve and determine (i) diversity factor (ii) units generated per day (iii) load factor.

(B) A chimney is 26 metres high and the temperature of hot gases inside the chimney is 350°C . The temperature of outside air is 25°C and furnace is supplied with 13 kg of air per kg of coal burnt. Calculate

(a) Draught in mm of water.

(b) Draught height in metres of hot gases.

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At the end of a power distribution system, a certain feeder supplies three distribution transformers, each one supplying a group of customers whose connected load are listed as follows :

Transformer 1
General power
service and lighting

Transformer 2
Residence lighting

Transformer 3
Store lighting and power

a : 20 H.P., 15kW

e : 12 kW

i : 20 kW

b : 13 H.P., 14kW

f : 12 kW

l : 16 kW

c : 11 H.P.

g : 6 kW

j : 22 kW, 7 H.P.

d : 7 H.P., 2 kW

h : 15 kW

k : 6 kW, 23 H.P.

Demand factor for all type of load is .62 & Diversity factor as 1.35

Use the factor and predict the maximum demand on the feeder. The H.P. load is motor load and assume an efficiency of 75%.

15.(A)

A power station has to meet the following demand :

Group A : 150 kW between 8 A.M. and 6 P.M.

Group B : 350 kW between 6 A.M. and 10 A.M.

Group C : 350 kW between 6 A.M. and 10 A.M.

Group D : 50 kW between 10 A.M. and 6 P.M. and then between 6 P.M. and 6

A.M.

Plot the daily load curve and determine (i) diversity factor (ii) units generated per day (iii) load factor.

(B) A chimney is 29 metres high and the temperature of hot gases inside the chimney is 320°C . The temperature of outside air is 27°C and furnace is supplied with 15 kg of air per kg of coal burnt. Calculate

(a) Draught in mm of water.

(b) Draught height in metres of hot gases.

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At the end of a power distribution system, a certain feeder supplies three distribution transformers, each one supplying a group of customers whose connected load are listed as follows :

Transformer 1
General power
service and lighting

Transformer 2
Residence lighting

Transformer 3
Store lighting and power

a : 22 H.P., 14kW

e : 15 kW

i : 25 kW

b : 13 H.P., 14kW

f : 12 kW

l : 16 kW

c : 14 H.P.

g : 6 kW

j : 28 kW, 7 H.P.

d : 7 H.P., 12 kW

h : 12 kW

k : 16 kW, 23 H.P.

Demand factor for all type of load is .62 & Diversity factor as 1.35

Use the factor and predict the maximum demand on the feeder. The H.P. load is motor load and assume an efficiency of 75%.

17.(a)

The maximum (peak) load on a thermal power plant of 60 MW capacity is 50 mW at an annual load factor of 50%. The loads having maximum demands of 25 MW, 20 MW, 8 MW and, 5 MW are connected to the power station.

Determine: (a) Average load on power station (b) Energy generated per year (c) Demand factor (d) Diversity factor.

17(b)

In a steam power plant the capital cost of power generation equipment is Rs. 25×10^6 . The useful life of the plant is 30 years and salvage value of the plant to Rs. 1×10^5 . Determine by sinking fund method the amount to be saved annually for replacement if the rate of annual compound interest is 6%.

18

(a)

Two lamps are to be compared:

(a) Cost of first lamp is Re. 1 and it takes 100 watts.

(b) Cost of second lamp is Rs. 4 and it takes 60 watts.

Both lamps are of equal candle power and each has a useful life of 100 hours. Which lamp will prove economical if the energy is charged at Rs. 70 per kW of maximum demand per year plus 5 paise per kWh? At what load factor both the lamps will be equally advantageous?

(b) The input-output curve of a 150 MW station is expressed by the formula,

$$I = 10^6 (250 + 5L + 0.03L^3) + L^4$$

where I is in kcal per hour and L is in mega-watts. Find the load at which minimum heat rate occurs and check with the plot.

19 (A)

A power station has to supply load as follows:

Time (hours)	0-6	6-12	12-14	14-18	18-24
Load (MW)	30	90	60	100	50

(a) Draw the load curve. (b) Draw the load duration curve. (c) Select suitable generating units to supply the load. (d) Calculate the load factor. (e) Calculate the capacity of the plant and the plant capacity factor.

(B) The capital cost of a power generating equipment in a steam power plant is Rs 80×10^6 . The useful life of the plant is 30 years and its salvage value is 7% of the capital cost. Determine by the sinking-fund method the amount of money to be saved annually for replacement if

the yearly rate of compound interest is 6.5%.

20 (a)

The annual peak load on a 30 MW power station is 25 mW. The power station supplies load having maximum demands of 10 MW, 8.5 MW, 5 MW and 4.5 MW. The annual load factor is 0.45. Find:

- (i) Average load
- (ii) Energy supplied per year
- (iii) Diversity factor
- (iv) Demand factor

(b) In a steam power plant the capital cost of power generation equipment is Rs. 35×10^5 . The useful life of the plant is 25 years and salvage value of the plant is Rs. 2×10^5 . Determine by sinking fund method the amount to be saved annually for replacement if the rate of annual compound interest is 5%.

21 (A)

(A power station has a maximum demand of 15 MW, a load factor of 0.7, a plant capacity factor of 0.525 and a plant use factor of 0.85. Find:

- (i) The daily energy produced.
- (ii) The reserve capacity of the plant.
- (iii) The maximum energy that could be produced daily if the plant operating schedule is fully loaded when in operation.

(B) The input output curve of a 10MW power station is expressed as follows:

$$i = 10^6(8 + L + 0.4L^2)$$

where i is in kcal per hour and L is in MW.

Determine the average heat rate of the power station for a day if it operates at its full capacity for 12 hours and is kept running at zero load for remaining 12 hours.

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(a) A residential load of a locality is given below:

Time (h)	0-5	5-6	6-9	9-18	18-21	21-24
Load (kW)	2	6	18	0	12	8

Draw the load curve and find out the load factor and energy consumed during 24 hr.

(b) A power plant has the following annual factors: Load factor = 70 per cent; capacity factor = 50 per cent. Use factor = 60 per cent. Maximum demand is 20 MW. Find out the following:

- (a) Annual energy production
- (b) Reserve capacity over and above peak load
- (c) Hours during which the plant is not in service per year

23 (a)

A power station supplies the following loads to the consumers:

Time (h)	0-6	6-10	10-12	12-16	16-20	20-22	22-24
Load (MW)	30	50	60	40	80	70	40

Find (a) the load factor of the plant and (b) what is the load factor of a standby equipment of 30 MW capacity if it takes up all loads above 60 MW?

(b) In a steam power plant the capital cost of power generation equipment is Rs. 15×10^6 . The useful life of the plant is 30 years and salvage value of the plant to Rs. 1.25×10^5 . Determine by sinking fund method the amount to be saved annually for replacement if the rate of annual compound interest is 4%.

24 (a)

The peak load on a power station is 35 MW. The loads having maximum demands of 15, 10, 5 and 7 MW are connected to the power station. The capacity of the power station is 40 MW and annual load factor is 50 per cent. Find (a) the average load on the power station, (b) energy supplied per year, (c) demand factor and (d) diversity factor.

(b) The input-output curve of a 150 MW station is expressed by the formula,

$$I = 10^6 (250 + 5L + 0.03L^3) + L^4$$

where I is in kcal per hour and L is in mega-watts. Find the load at which minimum heat rate occurs .

25 (a)

A power plant supplies the loads having maximum demands of 40, 50 and 30 MW respectively. The load factor of the plant on the basis of annual load curve is 60 per cent and the diversity factor of the load is 1.2.

Determine (a) the maximum load on the power plant, (b) the capacity of the power plant required to take the loads and (c) annual energy supplied by the power plant.

(b) The capital cost of a power generating equipment in a steam power plant is Rs 70×10^6 . The useful life of the plant is 20 years and its salvage value is 8% of the capital cost. Determine by the sinking-fund method the amount of money to be saved annually for replacement if the yearly rate of compound interest is 5%.

26 (A)

A power plant supplies the loads having maximum demands of 40, 50 and 30 MW respectively. The load factor of the plant on the basis of annual load curve is 60 percent and the diversity factor of the load is 1.2.

Determine (a) the maximum load on the power plant, (b) the capacity of the power plant required to take the loads and (c) annual energy supplied by the power plant.

(B) The following loads are connected to a power plant:

Type of load	Max. demand (MW)	Diversity factor	Demand factor
Domestic	15	1.5	0.75
Commercial	25	1.25	0.85
Industrial	50	1.35	0.9

If the overall diversity factor is 1.5, determine (a) the maximum load, and (b) the connected load of each type.

27(A) The annual peak load on a 35 MW power station is 30 MW. The power station supplies load having maximum demands of 9MW, 12 MW, 5 MW and 5.5 MW. The annual load factor is 0.55 Find:

(i) Average load

- (ii) Energy supplied per year
- (iii) Diversity factor
- (iv) Demand factor

(B) The input-output curve of a 150 MW station is expressed by the formula,

$$I = 10^6 (300 + 6L + 0.035L^3) + L^4$$

where I is in kcal per hour and L is in mega-watts. Find the load at which minimum heat rate occurs and check with the plot.

28(A)

A load duration curve for an industrial load is served by hydro-thermal plants follows a straight line. The maximum and minimum loads are 30 and 10 MW, respectively. The hydel-plant is available at the time of minimum regulating flow which is just enough to take a peak load of 75 MWh per day. On investigation, it was found economical to pump water from lower reservoir to higher reservoir using the excess power from steam plant during peak period allowing steam plant to run at 100 per cent load all the times. Determine the capacities of hydel and steam plants required for the purpose. Assume efficiency of steam to electric conversion = 45%

(B) In a steam power plant the capital cost of power generation equipment is Rs. 30×10^5 . The useful life of the plant is 27 years and salvage value of the plant to Rs. 1×10^5 . Determine by sinking fund method the amount to be saved annually for replacement if the rate of annual compound interest is 5%.

29(A)

The yearly duration curve for a certain plant can be considered as a straight line from 20 to 5 MW, WHERE power IS supplied BY one generator rated at 5 MW and two GENERATING units of 10 MW capacity each. Determine

- (a) Installed capacity
- (b) Plant factor
- (c) Maximum demand
- (d) Load factor
- (e) Utilization factor

(B) A central power station has annual load factor, capacity factor and use factor

as 60%, 40% and 45%, respectively. The maximum demand is 15 MW. Estimate (a) the annual energy production and revenue earned if the cost of energy is Re 1/UNIT, (b) the reserve capacity over and above the peak load, and (c) the hours per year the station is not in service.

30 (A)

A power plant has the following annual factors: Load factor = 70 per cent; capacity factor = 50 per cent. Use factor = 60 per cent. Maximum demand is 20 MW. Find out the following:

- (a) Annual energy production
- (b) Reserve capacity over and above peak load
- (c) Hours during which the plant is not in service per year

(B) The incremental fuel costs for two generating units A and B of a power plant are given by the following relations:

$$dF_A/dP_A = 0.07 P_A + 10$$

$$dF_B/dP_B = 0.08 P_B + 12$$

where P is the power in MW and F is the fuel cost in rupees per hour.

- (a) Find the economic loading of the two units when the total load to be supplied by the power station is 120 MW.

Find the net increase in fuel cost per hour if the load is equally shared by the two units.

31 (A)

A base load power station and a standby power station share a common load as follows:

Base load station annual output = 150×10^6 kWh

Base load station capacity = 25 MW

Maximum demand on base load station = 30 MW

Stand by station capacity = 18 MW

Maximum demand on standby station = 15 MW

Stand-by-station annual output = 14×10^6 kWh

Maximum demand on standby station = 15 MW

Determine the following for both power stations: (a) load factor and (b) capacity factor

(B) The capital cost of a power generating equipment in a steam power plant is Rs 90×10^6 . The useful life of the plant is 25 years and its salvage value is 10% of the capital cost. Determine by the sinking-fund method the amount of money to be saved annually for replacement if the yearly rate of compound interest is

3%.

32(A)

The peak load on a power station is 40 MW. The loads having maximum demands of 18, 12, 8 and 9 MW are connected to the power station. The capacity of the power station is 50 MW, annual load factor is 62 per cent find,

- (i) Average load on the power station
- (ii) Energy supplied per year
- (iii) Demand factor
- (iv) Diversity factor

(B) A power plant has the following annual factors: Load factor = 70%; Capacity factor = 50%; Use factor = 60%; Maximum demand = 20 MW. Estimate (a) annual energy production, (b) reserve capacity over and above the peak load, and (c) the hours per year during which the plant is not in service.

33 (A)

(The annual peak load on a 30 MW power station is 25 MW. The power station supplies load having maximum demands of 10 MW, 8.5 MW, 5 MW and 4.5 MW. The annual load factor is 0.5. Find:

- (i) Average load
- (ii) Energy supplied per year
- (iii) Diversity factor
- (iv) Demand factor

(B) The input-output curve of a 150 MW station is expressed by the formula,

$$I = 10^6 (250 + 5L + 0.03L^3) + L^4$$

where I is in kcal per hour and L is in mega-watts. Find the load at which minimum heat rate occurs and check with the plot.

34 (A)

The maximum demand of a power station is 26,000 kW and daily load curve is described as follows:

Time (h)	0-4	4-8	8-12	12-16	16-20	22-24
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Load (MW)	8	15	28	15	42	8
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(i) Determine the load factor

(ii) What is the load factor of stand by equipment rated at 8 MW that takes up all load in excess of 15 MW. Calculate use factor also.

(B) The capital cost of a power generating equipment in a steam power plant is Rs 80×10^6 . The useful life of the plant is 30 years and its salvage value is 5% of the capital cost. Determine by the sinking-fund method the amount of money to be saved annually for replacement if the yearly rate of compound interest is 6%.

35 (A)

Draw load curve and the load duration curve from the following data. Find the load factor and utilization factor of the plant serving this load if its capacity is 90 MW.

Time	12 PM	2 AM	6 AM	8 AM	12 AM	12.30 PM	1 PM	5 PM	6 PM	12 PM
Load (MW)	18	8	8	48	48	38	48	48	68	18

(B)

The connected loads of a consumer consist of 10 electric bulbs of 60 W each and 2 electric heaters of 1000 W each. His maximum demand is 1500 W. On an average, he uses 8 bulbs for 5 hours a day and one heater for 3 hours a day. Estimate his average load, monthly energy consumption and load factor.

36 (A)

A power station has a maximum demand of 80×10^3 kW and daily load curve is defined as follows:

Time (h)	0-6	6-8	8-12	12-14	14-18	18-22	22-24
Load	40	50	60	50	70	80	40

(MW)							
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(a) Draw load curve and the load duration curve .Determine the load factor of power station.

(b) What is the load factor of standby equipment raid at 25 MW that takes up all load in excess of 60 MW? Also calculate its use factor.

(B)An yearly load duration curve of a gas turbine power plant is a straight line from 50,000 to 5,000 kW. The load is taken by a power plant that consists of two units of 30,000 kW each and one unit of 20,000 kW. Determine (a) load factor and (b) capacity factor of the plant.

37 (a)

A power station has two 60 MW unit each running for 7,000 hr a year and an 30 MW unit running for 1,500 hr a year. The energy produced per year is 700×10^6 kWh, calculate the following:

(a) Plant load factor

(b) Plant use factor

(b)The original value of an equipment is Rs2,50,000 and its salvage value at the end of its useful life of 20 years is Rs25,000. Find the value of the equipment at the end of 10 years of its use by the following methods: (a) straight line depreciation and (b) sinking fund depreciation, when it is compounded annually at 8%.

38 (A)

The loads on a power plant with respect to time for 24 hr are listed below:

Time (h)	0-6	6-8	8-12	12-14	14-18	18-22	22-24
Load (MW)	40	40	60	50	70	80	40

Draw the load curve,LOAD DURATION CURVE and find cut the load factor of the power station. If the loads above 60 MW are taken by stand by unit of 20 MW capacity, find out the load factor of the stand by unit.

(B)A power station has two 60 MW unit each running for 7,000 hr a year and an 30 MW unit running for 1,500 hr a year. The energy produced per year is $700 \times$

10 6 kWh, calculate the following:

- (a) Plant load factor
- (b) Plant use factor

39 (A)

The daily load of a power station is given below:

Time	12 night	12 AM	6 AM	8 AM	12 noon	12:30 PM	1 PM	5 PM	6 PM	12 night
Load (MW)	20	10	10	40	40	30	40	40	60	20

- (a) Plot load and load duration curve for the system
- (b) Find out the load factor

(B)The incremental fuel costs of two generating units A and B of a power station are given by the following expressions:

$$dF_A/dP_A = 0.025 P_A + 2.5$$

$$dF_B/dP_B = 0.02P_B + 1.75$$

where F is in rupees per hour and P it is MW. Determine incremental fuel cost and loading schedule for minimum cost if the total load to be supplied is to be 50 MW 175 MW and 200MW. Both units operate at all times and maximum and minimum load on each unit is to be 100 MW and 15 MW respectively.

40(A)

The input output curve of a 10MW power station is expressed as follows:

$$i = 10^6(8 + L + 0.4L^2)$$

where i is in kcal per hour and L is in MW.

Determine the average heat rate of the power station for a day if it operates at its full capacity for 12 hours and is kept running at zero load for remaining 12 hours. Also calculate the saving per kWh of energy produced if the energy is generated at a constant 24 hours load (100% load factor).

(B)

The following data are collected from a daily load curve of a power plant.

Load (kW)	15000	12000	10000	8000	6000	4000	2000
No of hours at	875	876	1750	2630	4380	7000	8740

load							
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(a) Find out the load factor of the system.

(b) If the load is supplied by two plants one is acting as base load plant of 10,000 kW capacity and other as peak load plant of 7,500 kW capacity, find out the annual load factor, annual capacity factor and annual use factor for both plants.

41(A)

A power station has a maximum demand of 80 x 10 k W and daily load curve is defined as follows:

Time (h)	0-6	6-8	8-12	12-14	14-18	18-26	22-24
Load (MW)	40	50	60	50	70	80	40

(a) Draw the load CURVE AND load duration curve and find out the load factor of the power station.

(b) What is the load factor of standby equipment rated at 25 MW that takes up all load in excess of 60 MW ? Also calculate its use factor.

(B) A chimney is 28 metres high and the temperature of hot gases inside the chimney is 320°C. The temperature of outside air is 27°C and furnace is supplied with 15 kg of air per kg of coal burnt. Calculate

(a) Draught in mm of water.

(b) Draught height in metres of hot gases.

42 (A)

Find the diversity factor of power station which supplies the following loads:

Load A: Motor load of 100 kW working between 10AM. and 6P.M.

Load B: Lighting load of 60 kW between 6 P.M. and 10A.M.

Load C: Pumping load of 40 kW between 4 P.M. and 10AM.

(B)

In a steam power plant the capital cost of power generation equipment is Rs. 30×10^5 . The useful life of the plant is 24 years and salvage value of the plant is Rs. 1.5×10^5 . Determine by sinking fund method the amount to be saved annually for replacement if the rate of annual compound interest is 7%.

44 (A)

The annual peak load on a 30 MW power station is 25 MW. The power station supplies loads having maximum demands of 10MW, 8.5 MW, 5 MW and 4.5 MW. The annual load factor is 45%.

Find:

- (a) Average load,
- (b) Energy supplied per year,
- (c) Diversity factor,
- (d) Demand factor.

(B) Determine the height of chimney to get net draught of 12 mm if the total draft losses are 4mm. The temperature of air is 25°C and the temperature of chimney gases is 300°C. The mass of air used per kg of fuel used is 18 kg. One kg of air occupies a volume of 0.7734 m³ at N.T.P.

45(A)

For a power station the yearly load duration curve is a straight line from 30,000 to 4,000 kW. To meet the load three turbo-generators are installed. The capacity of two generators is 15,000 kW each and the third is rated at 5,000 kW. Determine the following:

- (a) Load factor,
- (b) Capacity factor or plant factor,
- (c) Maximum demand.

(B) The daily load of a power station is given below:

Time	12 night	12 AM	6 AM	8 AM	12 noon	12:30 PM	1 PM	5 PM	6 PM	12 night
Load (MW)	20	10	10	40	40	30	40	40	60	20

- (a) Plot load and load duration curve for the system
- (b) Find out the load factor

46(A)

The maximum (peak) load on a thermal power plant of 60 MW capacity is 50 MW at an annual load factor of 50%. The loads having maximum demands of 25 MW, 20 MW, 8 MW and 5 M are connected to the power station. Determine : (a) Average load on power station (b) Energy generated per year (c) Demand factor (d) Diversity factor.

(B) The capital cost of a power generating equipment in a steam power plant is Rs 80×10^6 . The useful life of the plant is 30 years and its salvage value is 5% of the capital cost. Determine by the sinking-fund method the amount of money to be saved annually for replacement if the yearly rate of compound interest is 6%.

47 (A)

A power station is to supply three region of load whose peak loads are 20 MW, 15 MW and 25 MW. The annual load factor is 50% and the diversity factor of the load at the station is 1.5. Determine the following:

- (a) Maximum demand on the station,
- (b) Installed capacity suggesting number of unit.
- (c) Annual energy supplied.

(B) Determine the height of chimney to get net draught of 14 mm if the total draft losses are 5 mm. The temperature of air is 30°C and the temperature of chimney gases is 20°C . The mass of air used per kg of fuel used is 18 kg. One kg of air occupies a volume of 0.7734 m^3 at N.T.P.

48 (A)

The load of a residential consumer for a day was found to be as follows:

From 12.00 midnight to 6A.M.	No load
From 6A.M. to 8A.M.	120W
From 8A.M. to 9.30A.M.	540 W
From 9.30A.M. to 11.30A.M.	No load
From 11.30A.M. to 2P.M.	480W
From 2 P.M. to 3.30 P.M.	No load
From 3.30 P.M. to 5P.M.	120 W
From 5 P.M. to 6.30 P.M.	360 W
From 6.30 A.M. to 9A.M.	480 W
From 9P.M. to 12 midnight	120 W

Calculate

- (a) Energy consumed
- (b) Load factor.

(B) A plant costing Rs 175,000 has a useful life of 120 years. Find the amount which should be saved annually to replace the equipment at the end of that time (a) by the straight-line method, and (b) by the sinking-fund method if the annual rate of compound interest is 5%. Assume that the salvage value of the equipment is Rs 50,000.

49 (A)

An input-output curve of a 10 MW station is expressed as follows:

$$I = 10^6(I_0 + 8L + 0.4L^2)$$

where I is in kcal per hour and L is in mega watts.

(i) Without plotting any curve find the load at which the maximum efficiency occurs.

(ii) Find the increase in input required to increase station output from 3 to 5 MW by means of the input-output curve and also by incremental rate curve.

(B) The capital cost of a power generating equipment in a steam power plant is Rs 80×10^6 . The useful life of the plant is 30 years and its salvage value is 5% of the capital cost. Determine by the sinking-fund method the amount of money to be saved annually for replacement if the yearly rate of compound interest is 6%.